

TRANSPORTATION SIMPLEX METHOD

General idea: write the problem in max form
 add slack variables
 select initial BFS
 start pivoting

BV	Eq	Z	x_{ij}	z_i	z_{m+j}	RHS
Z	(0)	-1	c_{ij}	π	π	0
	(1)	...	...	...	...	
z_i	(i)	0	1	1	0	s_i
	...					
z_{m+j}	(m+j)	0	1	0	1	d_j
	...					
	(m+n)	...	...	...	...	

x_{ij} DECISION VARIABLE (i source, j destination)

z_i SLACK VARIABLE for SUPPLY CONSTRAINTS

z_{m+j} SLACK VARIABLE for DEMAND CONSTRAINTS

m # SOURCES

n # DESTINATIONS

After any iteration, the table will look like

BV	Eq	Z	x_{ij}	z_i	z_{m+j}	RHS
Z	(0)	-1	$c_{ij} - u_i - v_j$	$\pi - u_i$	$\pi - v_j$	$-\sum_{i=1}^m s_i u_i - \sum_{j=1}^n d_j v_j$

u_i = MULTIPLE of ORIGINAL row (i) that HAS BEEN SUBTRACTED from the ORIGINAL row (0) DURING THE ITERATIONS of the SIMPLEX METHOD

v_j = MULTIPLE of original row (m+j) that HAS BEEN SUBTRACTED from the ORIGINAL row (0) DURING the ITERATIONS of the SIMPLEX METHOD

(if you have seen DUALITY, these are the DUAL VARIABLES of the SUPPLY and DEMAND CONSTRAINTS, respectively)

$c_{ij} - u_i - v_j \sim$ RATE at which Z will CHANGE AS x_{ij} IS INCREASED

- NEEDED INFO:
- INITIAL BFS (obtained adding variables as basic variables)
 - Row (0) (obtained by adding/subtracting other rows)
 $\Rightarrow$ BV ENTERING BV
 - COEFFICIENTS in "CONSTRAINTS" rows
 $\Rightarrow$ BV LEAVING BV
 - Row reduction and GAUSSIAN ELIMINATION

"TRANSITION" to TRANSP: no artificial variables needed to construct BFS

- Row (0) can be obtained without other rows by calculating u_i and v_j
 $\Rightarrow$ BV must have 0 coeff
 $c_{ij} - u_i - v_j = 0$ if x_{ij} is BV
- LEAVING BV can be found without knowing the coeff of ENTERING BV

SOURCE	DESTINATION				SUPPLY	u_i
	1	2		n		
1	c_{11}	c_{12}		c_{1n}	s_1	
2	c_{21}	c_{22}		c_{2n}	s_2	
m	c_{m1}	c_{m2}		c_{mn}	s_m	
DEMAND	d_1	d_2		d_n		
v_j						

c_{ij}	
	x_{ij}
BV	

c_{ij}	
	$c_{ij} - u_i - v_j$
non-BV	

INITIALIZATION: to obtain initial BFS

Any supply CONSTR
EXACTLY EQUALS SUM
OF DEMAND CONSTR
MINUS SUM OF OTHER
SUPPLY CONSTR
(SAME FOR DEMAND)

REMARK: there are $m+n$ functional constraints
 $\Rightarrow$ there are $m+n-1$ basic variables

(one constraint is redundant since
there are only equality constraints)

only BFS set appears in the Transportation
Simplex table with $m+n-1$ circled non-negative
allocations where the sum of the allocations
for each row or column equal its supply or
demand, respectively.

To construct an initial BFS we select
 $m+n-1$ basic variables (one at a time)

After each selection, a value ~~that~~ that satisfy
an additional constraint is assigned to
that ~~value~~ variable

$\Rightarrow$ after $m+n-1$ selections an entire
basic solution is constructed and it
satisfy all the constraints

GENERAL PROCEDURE for INITIAL BFS

0. all rows and columns are under consideration
at the beginning

1. From rows and columns under consideration,
select the next basic variable according to
some criterion (NORTHWEST CORNER RULE)

2. Make basic variable large enough to use all
the remaining supply in its row or the
remaining demand in its column
(the smallest of the two)

3. Eliminate the row or column from further
consideration (the one with supply or
demand satisfied)

If supply = demand, eliminate the row

4. If only one row or column remains under consideration, the procedure is completed by rekeying every remaining variable associated to that row or column to be basic with the only feasible allocation

OPTIMALITY TEST: A BFS is optimal iff $c_{ij} - u_i - v_j \geq 0 \quad \forall i, j$ such that x_{ij} is non basic

=> how do we compute u_i and v_j ?

- If x_{ij} is a basic variable (circled one) then $c_{ij} - u_i - v_j = 0$ (Zero coeff in row(0))

- There are $m+n-1$ basic variables but

$m+n$ u_i and v_j so one is redundant

=> select u_i that has the largest number of allocations in its row and assign to it 0 (Zero)

ITERATION 1. FIND ENTERING BASIC VARIABLE

$c_{ij} - u_i - v_j$ represent the rate at which the objective function changes (coeff of row(0))

=> the ENTERING BV must have the most negative

$c_{ij} - u_i - v_j$ to decrease the cost

2. FIND LEAVING BASIC VARIABLE

Increasing the entering basic variable set off a chain of reaction of compensating changes in other basic variables

=> the first that decreases to 0 is the LEAVING BV

- We have RECIPIENT cells that receive additional allocation and DONOR cells that "give" allocation

=> they all correspond to basic variables of the BFS (except the entering BV)

=> the DONOR cell with the smallest allocation provides the LEAVING BV

3. FIND NEW BES

By adding the value of the leaving BV to the allocation for each recipient cell and subtracting this same value from the allocation of each donor cell